CS-571 — Computer Vision Complex Computing Problem

Autonomous Visual Intelligence System (alternative implementation)

---

1. Executive Summary

This system builds an outdoor-scene understanding pipeline from four stages. Edges and lines are found with Gaussian smoothing, a median-based automatic Canny threshold, and both standard and probabilistic Hough transforms. Cross-view correspondence uses SIFT with a brute-force matcher, Lowe's ratio test and homography-based geometric verification. A panorama is built manually by projecting frames onto a cylinder, aligning them with RANSAC homographies and feather-blending them. Finally a pretrained MobileNetV2 is fine-tuned on EuroSAT land-use imagery, reaching a best validation accuracy of 96.2% (target 70%).

2. Pipeline Architecture



Figure 1 — five-stage data flow (geometry stages feed the learned stage).

Headline: T2 keypoints A/B=3033/3421, good matches=1825, homography inliers=1814 | T3 mean inlier ratio=70.8% | T4 best acc=96.2%

3. Quantitative Results

T1 — Hough experiments

Task 1 — Hough experiments

image	hough	params	#lines
house	standard	140	14
house	probabilistic	45/45/12	111
street	standard	140	9
street	probabilistic	45/45/12	16

T2 — SIFT stats

Task 2 — SIFT matching statistics

metric	value
keypoints A	3033
keypoints B	3421
good matches (ratio 0.75)	1825
cross-check matches	2040
homography inliers	1814
match quality ratio	0.602

T3 — RANSAC inliers

Task 3 — pairwise stats

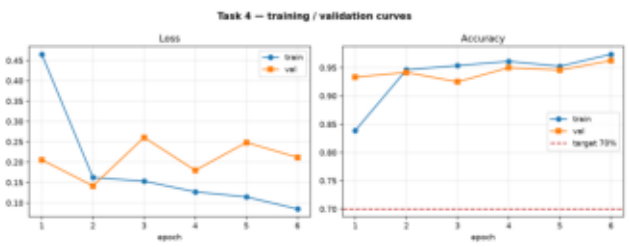
pair	#matches	RANSAC inliers
1-2	133	63.2%
2-3	98	74.5%
3-4	130	74.6%

T4 — summary

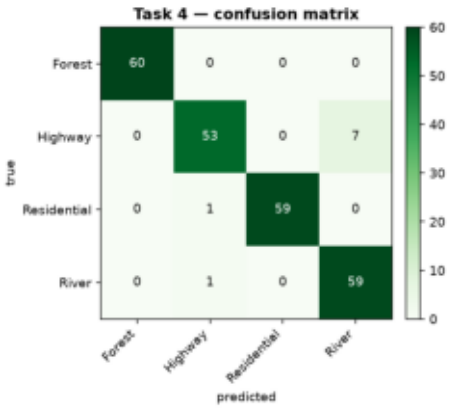
Task 4 — summary

metric	value
backbone	mobilenet_v2
dataset	EuroSAT (land-use)
classes	Forest, Highway, Residential, Rive
images/class (train)	200
best val accuracy	0.963
target 70% met?	YES
Grad-CAM samples	3

T4 — curves



T4 — confusion



T4 — per-class

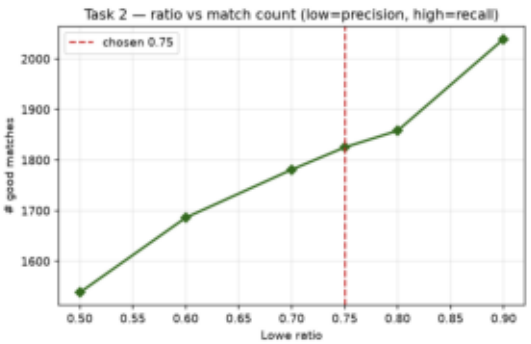
Task 4 — per-class metrics

class	precision	recall	F1	support
Forest	1.00	1.00	1.00	60
Highway	0.96	0.88	0.92	60
Residential	1.00	0.98	0.99	60
River	0.89	0.98	0.94	60

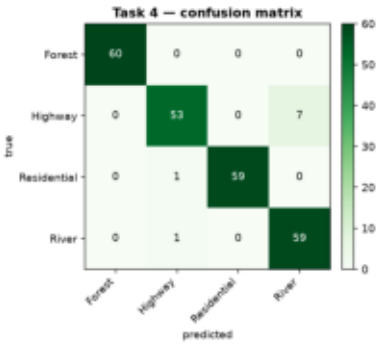
4. Failure Analysis (>= 2 per task)

Task 1. (a) A median-based auto threshold adapts to global brightness but still over-responds to fine texture, adding spurious short Hough segments. (b) The standard Hough returns infinite lines that ignore segment endpoints, so it merges distinct co-linear structures; the probabilistic variant is preferred for real edges.
Task 2. (a) Repetitive / self-similar texture yields ambiguous descriptors and wrong matches that only geometric (homography) verification removes. (b) Ratio 0.9 admits false positives (low precision); 0.5 starves recall — quantified by the ratio sweep.
Task 3. (a) Too little overlap makes RANSAC homography unstable (flagged as an edge case). (b) Wrong cylindrical focal length under/over-curves frames, leaving residual seams that feathering only partly hides.
Task 4. (a) Spectrally similar land-use classes (e.g. River vs Highway) are confused — see the off-diagonal confusion cells. (b) The Grad-CAM incorrect sample shows the net focusing on texture/background rather than the class-defining region.

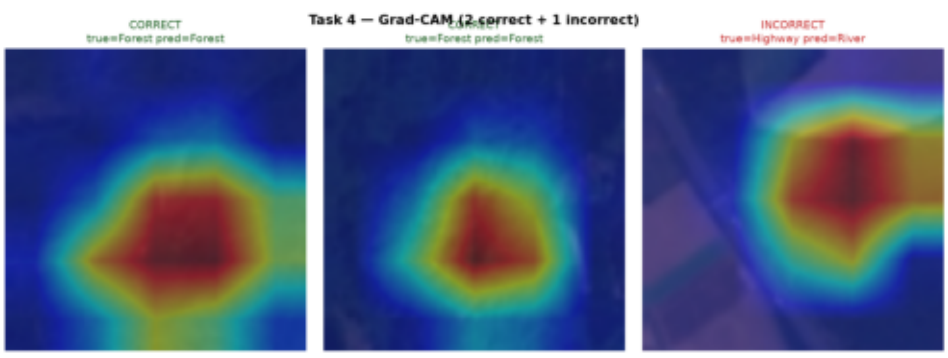
T2 — ratio sweep



T4 — confusion



T4 — Grad-CAM



5. Classical vs. Deep-Learning Trade-offs

Dimension	Classical (T1-T3)	Deep learning (T4)
Speed	Real-time on CPU	GPU/MPS to train
Accuracy	Exact under valid geometry	High semantics, distribution-sensitive
Interpretability	Transparent maths	Opaque; needs Grad-CAM
Data need	None (tuned params)	>=100 labelled images/class
Failure	Low texture / overlap	Similar-class confusion

6. Limitations & Future Work

- Mosaic frames are overlapping crops of a wide scene; genuine multi-camera capture with parallax would need bundle adjustment.
- EuroSAT is satellite land-use; a ground-level dataset would better match a driving robot, and the full taxonomy would test generalisation.
- SIFT + brute-force matching is CPU-bound; a learned detector or GPU matcher would raise throughput.
- Future work: fuse the geometric mosaic with the semantic labels into one scene map and add temporal consistency.

Reproducible via 'python main.py' (fixed seed). Data cited in README (OpenCV samples, BSD-3; EuroSAT land-use).